

Object Segmentation to prevent contextual shortcut learning

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Abstract

This project will focus on image classification using a deep learning model trained on a 10-class subset of the iNaturalist-12k dataset available on Kaggle, provided by Aryan Pandey. The dataset contains 12,000 images across biological categories such as Amphibia, Mammalia, and Plantae. The classes are fairly balanced with 1200 available images for each of the classes. The main goal is to design, train, and evaluate a convolutional neural network from scratch on the aforementioned dataset. As a possible extension, the project may explore feature engineering techniques such as extraction of the biological object (animal or plant) to study the impact of background information on classification. Additional extensions include the creation of a second model by transfer learning and the comparison of the latter with the model created from scratch.

1 Introduction

Deep learning models achieve strong performance on standard benchmarks, yet their limitations remain an active area of research. Geirhos et al. define shortcuts as decision rules that perform well on benchmark datasets but fail under more challenging or out-of-distribution conditions [2]. Such shortcut learning is particularly problematic in safety-critical applications such as autonomous driving, where models must generalize to unseen environments [1]. At the same time, prior work has shown that image backgrounds can contain highly predictive information and may substantially contribute to classification accuracy [11].

This project investigates whether reducing background information through automatic object segmentation can improve robustness and reduce shortcut learning in biological image classification. A 10-class subset of the iNaturalist-12K dataset from Kaggle was used [6]. The dataset is well suited for this task because it contains substantial variability in habitats, lighting conditions, and objects. A segmentation pipeline based on Segment Anything (SAM) [5, 8] and CLIP [7] was developed to automatically extract the foreground object without requiring manual annotations. The resulting segmented masks were used to crop images to the region of interest and to train CNN-based classification models. The performance of models trained with this segmentation-based regularization approach was then compared against equivalent models trained on the original, non-segmented images.

2 Related work

Shortcut learning has emerged as an important limitation of deep neural networks [2]. In ecological datasets, background information and hence the location can become a strong shortcut because images from the same location often share very similar environmental context, particularly in datasets collected from fixed animal camera traps [1]. Geirhos et al. argue that shortcut learning should not only

be studied through in-distribution accuracy, but rather by manipulating the evidence available to the model, such as background, texture, or color, and observing how predictions change [2]. Several studies demonstrate that image backgrounds alone can contain sufficient information for classification [11, 12]. Xiao et al. showed that models trained on ImageNet often strongly depend on contextual background information, while Rosenfeld et al. demonstrated that even small background perturbations such as noise can significantly alter predictions [10, 11]. Geirhos et al. further found that CNNs frequently rely more on texture cues than on object shape, which may contribute to such behavior and provides additional motivation for segmentation-based approaches that emphasize object structure over contextual texture information [3].

These findings motivate approaches that explicitly separate foreground objects from their surrounding context. However, many supervised segmentation methods, including segmentation-based variants of YOLO, require manually annotated masks and labels for training, which is highly labor-intensive and not given for the iNaturalist dataset [9]. Such annotations are not available for the iNaturalist-12K dataset used in this project. However, Segment Anything (SAM) [5, 8] provides a way to automatically generate segmentation proposals without requiring manual annotation. To semantically identify relevant masks, this project additionally uses CLIP [7], which uses text-image similarity to semantically score and label images. The resulting pipeline is conceptually related to copy-paste augmentation methods [4], where segmented foreground objects are inserted into alternative backgrounds to improve robustness and reduce contextual bias.

3 Experiments

3.1 Dataset

The dataset used for this project is a 10-class subset of the iNaturalist-12k dataset available on Kaggle (link in appendix), provided by Aryan Pandey. It contains 12,000 images across biological categories such as Amphibia, Mammalia, and Plantae. The classes are balanced with 1,200 available images for each of the classes. The dataset was split into 8,000 training images, 2,000 validation images, and 2000 test images, stratified by class.

Model Architecture. Figure 1 shows the custom CNN used in this work. It consists of four convolutional blocks, each applying two 3×3 convolutions with batch normalization and ReLU, followed by 2×2 max pooling. Global average pooling and a small fully connected classifier produce the final 10-class logits. As a transfer-learning baseline, a pretrained ResNet18 with ImageNet weights was used by replacing its final fully connected layer with a new 10-class classifier.

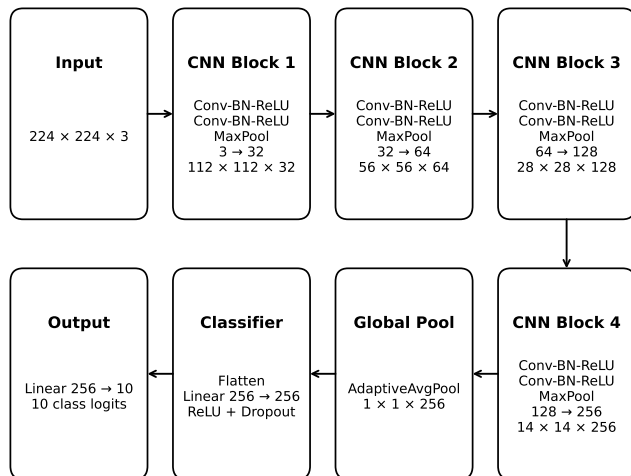


Figure 1: Architecture of custom model

3.2 Training Details

Models were trained with cross-entropy loss, AdamW, batch size 32, image size 224×224, random resized crops, horizontal flips, dropout 0.3, weight decay, and gradient clipping at 1.0. Early stopping was used during model selection. Final training used the selected epoch counts without early stopping: 122 epochs for the custom CNN and 30 for ResNet18. For the custom CNN, the final setting used learning rate 0.001 and weight decay 3e-4. ResNet18 used learning rate 0.001 and weight decay 1e-4. Each architecture was finally trained with segmentation probabilities $p \in \{0.0, 0.25, 1.0\}$, where p denotes the probability of replacing an original training image by its segmented version.

3.3 Results

The quality of the segmentation masks was manually evaluated on a sample of 200 masks, with 20 masks per class. A mask was counted as good if it captured the main foreground object without substantial missing regions or excessive background. The results in table 2 (appendix) show significant variation between classes. In particular, masks were weak for *Animalia*, but very reliable for *Aves*.

Table 1: Final test results for ResNet18 and the custom CNN under different segmentation probabilities.

Architecture	Seg. prob.	AUC	Recall	Precision
ResNet18	0.00	0.963	0.740	0.741
ResNet18	0.25	0.965	0.746	0.750
ResNet18	1.00	0.944	0.674	0.680
Custom CNN	0.00	0.887	0.544	0.554
Custom CNN	0.25	0.882	0.535	0.538
Custom CNN	1.00	0.697	0.241	0.278

Partial segmentation reduced overfitting during training, as the gap between training and validation loss, accuracy, precision, and

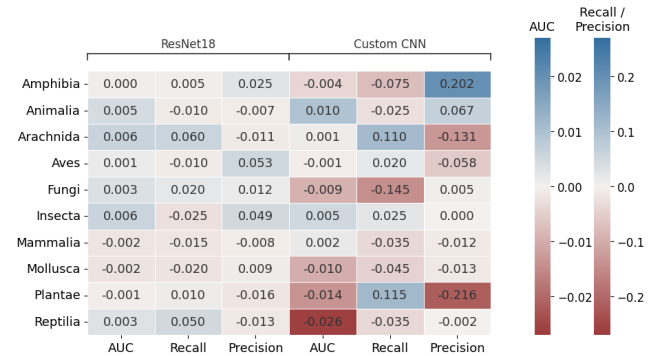


Figure 2: Comparison of models trained with partial segmentation ($p = 0.25$) against models trained on original images. Positive values indicate an improvement from segmentation; negative values indicate worse performance.

recall became smaller for $p = 0.2$ and $p = 0.5$. However, this came with slightly lower validation performance compared to the non-segmented baseline. Full segmentation ($p = 1.0$) performed poorly, which can also be observed in the final results of the final trained models in table 1. However, here, the performance of partial segmentation of 25% vs. no segmentation does seem to be almost equivalent in terms of performance, which indicates that it actually regularizes. Moreover, the ResNet18 model performs much better than the custom CNN.

Figure 2 shows the difference between the metrics between the models trained with 25% probability segmentation masks vs. non-segmented version for each class. In general, the ResNet did not improve or worsen by much (expected since pre-trained). Of particular interest was the comparison with respect to the *Avis* class, which seemed to be the best segmentation masks. However, not much of an improvement can be seen for the *avis* class. In general, there seems to be no global trend. It does improve some metrics from the custom CNN for a given class, but then another metric is worse, e.g. for *plantae* or *Arachnida* looking at precision vs. recall.

4 Discussion

- This experiment explored if segmentation regularization can actually help. During this setup, it was found that it does not improve or worsen the model by much, if applied as regularization (with probability 25%).
- Quality of masks not great. But even for seemingly good masks, it did not improve.
- Realm of study project quite limited, worth repeating under cleaner conditions with proper labelling. Talking about limitations: Commenting that this is just a student project, that the segmentation is very imperfect, that there were not enough resources to properly test all models. Not enough data either.
- Experiment worth to be better repeated for datasets with given segmentation masks. Maybe label them by company.
- Could go much further by putting e.g. a cow in the sky, which is related to what [3] and [10] found.

Table 2: Manual quality assessment of segmentation masks by class (n=200). Values indicate the percentage of masks rated as good.

Class	Amph.	Anim.	Arach.	Aves	Fungi
Good masks (%)	50	29	43	94	79
Class	Insecta	Mamm.	Moll.	Plant.	Rept.
Good masks (%)	74	81	74	76	67

- Outside of context, this experiment also showed that it is hard to beat a transfer-learned model in such a scenario, since there were only 12,000 images available, which is not so many.

5 References

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A Dataset

<https://www.kaggle.com/datasets/aryanpandey1109/inaturalist12k>

B Results

C Prompts for SAM

D Declaration of generative AI usage

I hereby declare that I have used an LLM (ChatGPT, model GPT-5.3 and GPT-5.5) to assist me with some small passages of the code, debugging, improve some readability, to get new ideas, to assist me with training on the HPC server, to visualize the model architecture, and to find better vocabulary and restructuring some sentences while writing the paper. This is in accordance with the "Guidelines on the Use of Generative AI for Teaching and Learning", Version: 1.0, Date: 2026-02-16.